

ARTIGAS BASE, Antarctica

WMO: 890540

Lat: 62.185S

Lon: 58.905W

Elev: 56

StdP: 14.67

Time Zone: -3.00 (W03)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.7	4.5	-3.3	4.6	2.0	0.4	5.6	5.2	54.1	9.9	46.4	19.2	14.9	100	1.098

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.7	40.2	38.1	39.2	37.5	38.4	37.0	38.7	39.4	38.1	38.7	37.5	38.0	17.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
38.2	33.9	38.9	37.4	32.9	38.2	36.7	32.0	37.6	14.8	39.6	14.5	38.9	14.2	38.2	44.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 41.6	34.8	31.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	1.9	40.5	3.5	1.8	-0.6	41.8	-2.7	42.9	-4.6	43.9	-7.1	45.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	35.1	35.3	33.2	30.0	27.5	23.4	21.5	22.5	24.1	27.7	30.4	32.9
	DBStd	7.01	1.93	2.27	3.30	5.17	5.17	7.35	7.66	7.63	6.49	4.30	2.93	2.55
	HDD50	7809	461	411	519	601	697	798	885	851	777	692	588	529
	HDD65	13284	926	831	984	1051	1162	1248	1350	1316	1227	1157	1038	994
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	15.9	14.6	15.1	15.2	16.4	15.8	16.4	16.0	16.8	16.5	17.9	15.8	14.7
	PrecAvg	40.9	3.2	3.2	4.7	4.0	3.8	3.0	2.8	3.0	3.2	3.9	3.3	2.8
(15) Precipitation	PrecMax	54.8	4.5	4.9	7.4	6.0	6.3	4.5	4.3	5.6	5.4	6.7	4.5	4.5
	PrecMin	33.7	1.9	1.9	2.8	2.3	1.7	1.6	1.3	0.6	1.1	2.0	1.2	1.2
	PrecStd	4.6	0.6	0.7	1.1	0.9	1.1	0.7	0.9	1.1	0.9	1.0	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	41.7	41.4	40.8	39.2	37.0	35.9	34.2	34.6	34.9	35.5	37.3	40.4	
	0.4% MCWB	39.0	39.5	38.3	37.8	35.4	34.0	33.1	33.6	33.7	34.3	36.1	37.6	
	2% DB	39.7	40.0	39.2	37.6	35.7	34.5	33.2	32.9	33.5	34.2	35.7	38.3	
	2% MCWB	37.7	38.4	37.9	36.5	34.6	33.3	32.4	32.0	32.5	33.2	34.6	36.4	
	5% DB	38.9	39.1	38.3	36.6	34.8	33.4	32.2	32.2	32.6	33.3	34.8	37.1	
	5% MCWB	37.1	37.6	37.2	35.7	33.9	32.6	31.6	31.4	31.9	32.5	33.8	35.5	
	10% DB	38.0	38.3	37.5	35.7	33.9	32.3	31.1	31.4	31.7	32.5	34.0	36.3	
	10% MCWB	36.5	37.1	36.5	34.9	33.1	31.7	30.6	30.8	31.2	31.8	33.2	34.8	
	0.4% WB	39.6	40.2	39.3	38.1	35.8	34.9	33.7	34.0	34.2	34.9	36.1	38.2	
	0.4% MCDB	40.6	40.8	39.9	38.9	36.2	35.5	33.8	34.5	34.5	35.2	36.5	39.2	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.6	38.7	38.3	36.9	34.9	33.7	32.6	32.3	32.9	33.6	34.9	36.9	
	2% MCDB	39.4	39.3	38.8	37.3	35.3	34.2	32.9	32.6	33.2	33.9	35.3	37.8	
	5% WB	37.7	38.2	37.6	35.9	34.1	32.8	31.6	31.6	32.1	32.8	34.1	36.0	
	5% MCDB	38.4	38.7	38.0	36.3	34.6	33.3	32.0	31.9	32.4	33.1	34.4	36.8	
	10% WB	36.9	37.5	36.8	35.0	33.2	31.7	30.6	30.9	31.3	32.1	33.4	35.1	
	10% MCDB	37.6	38.0	37.3	35.5	33.8	32.2	31.0	31.3	31.7	32.4	33.8	35.9	
	5% DB	4.1	3.7	4.4	4.7	5.6	6.5	7.7	6.9	6.7	5.0	4.3	3.9	
	5% DB MCDBR	5.5	4.6	4.8	4.9	5.2	6.1	6.4	5.6	5.7	4.9	4.8	5.2	
	5% DB MCWBR	4.0	4.0	4.1	4.7	4.9	5.7	6.6	5.8	5.6	4.5	4.1	4.3	
	5% WB MCDBR	4.9	4.1	4.5	4.8	4.7	5.7	6.0	5.8	5.7	4.7	4.2	5.1	
(35) Mean Daily Temperature Range	5% WB MCWBR	4.0	3.9	4.2	4.8	4.8	5.3	6.2	5.9	5.7	4.6	3.9	4.5	
	taub	0.309	0.308	0.297	0.280	0.225	0.366	0.211	0.268	0.288	0.310	0.304	0.309	
	taud	2.492	2.501	2.501	2.372	2.100	2.822	2.049	2.239	2.369	2.399	2.456	2.464	
	Ebn at Noon	307	291	267	214	141	96	139	210	263	286	306	310	
(42) Clear-Sky Solar Irradiance	Edh at Noon	31	28	23	17	10	6	10	18	25	30	32	33	
(44) All-Sky Solar Radiation	RadAvg	1587	1145	616	270	78	33	56	201	529	980	1501	1798	
	RadStd	146	88	51	22	7	1	3	17	42	65	91	143	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A	N/A

Nomenclature: See separate page